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Linked shrinkage prior: a Bayesian approach to detect interaction effects in survival analysis

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Summary

Graft loss in kidney transplantation involves complex interactions among the kidney donor, kidney recipient, and the transplantation process. To understand these interactions, previous hypothesis-driven studies have primarily focused on single interaction effects, often overlooking other two-way interactions. In this thesis, we adopt a data-driven approach to simultaneously investigate all two-way interactions. Building on the linked shrinkage prior proposed by van de Wiel et al., 2024, which connects interaction terms with their corresponding main effects, we apply this prior to Bayesian proportional hazards models using both full and partial likelihood estimators. For the baseline hazard function $h_0(t)$ in full likelihood estimator, we consider three scenarios: (1) a constant baseline hazard function, where true event times T^* follow an Exponential distribution; (2) a time-varying baseline hazard function, where T^* follow a Weibull distribution; and (3) no assumption about the baseline hazard function, where T^* do not follow a specific distribution. In the third case, we approximate the log-baseline hazard using B-Splines. The method is validated through simulations and applied to kidney transplantation data. Results indicate that the linked shrinkage prior improves the detection of interaction effects compared to a non-informative prior. However, its advantage diminishes when combined with full likelihood estimators and B-Spline approximation of the log-baseline hazard function in real-world kidney transplantation data. We conclude that the linked shrinkage prior is effective for detecting interaction terms in survival analysis, but careful handling of the baseline hazard estimation is crucial, with the partial likelihood estimator being a safer choice.

Introduction

In kidney transplantation research, graft loss refers to the time duration before a transplanted kidney fails. This outcome is typically modeled as a time-to-event variable and has traditionally been explained by average characteristics of the donor, recipient, and transplantation process, for instance, HLA mismatch, donor type (donation after circulatory death, DCD vs. donation after brain death, DBD), and donor sex (female vs. male). Recent findings by Steenvoorden, Evers, et al. (2024) report that interaction effects between donor type and donor sex may also play a important role in explaining graft loss. In their study, the authors began with a biological hypothesis and focused exclusively on the interaction between donor type and donor sex. All other interactions in their study were deliberately excluded. While this hypothesis-driven approach can increase statistical power for detecting the targeted interaction, it is inherently limited by its reliance on empirical theories, which may not always be comprehensive, accurate, or objective. As a result, interactions not anticipated by these theories are overlooked. As a complementary research approach, including all possible two-way interactions in time-to-event analyses may provide more comprehensive and/or objective insights into graft loss from a data-driven perspective,

especially into the joint effects of covariates. Methodologically speaking, including all possible two-way interactions can also help reduce bias in estimating main effects and improve model fit and/or predictive performance.

However, when including all possible two-way interactions in such kidney transplantation data (see details in the section *Detecting interaction effects in kidney transplantation*), where the sample size is limited relative to the number of covariates, three common problems may arise: (1) models such as Cox regression may fail to converge (see, e.g., Beca et al., 2021; Johnson et al., 1982); (2) issues related to multiple testing may occur (see, e.g., van de Wiel et al., 2024); and (3) large variance or standard errors of parameter estimates may arise. These challenges occur because (1) the number of interaction terms grows quadratically as the number of covariates increases; and (2) interaction terms are often highly correlated with their associated main effects.

In this thesis, we aim to address this gap by developing an interpretable and statistically robust approach that can accommodate a large number of parameters while still enabling valid parameter inference. Specifically, we focus on the accurate estimation and selection of interaction effects to better explain graft loss in kidney transplantation.

Background

Before addressing the accurate estimation and selection of interaction effects to explain graft loss in kidney transplantation, it is helpful to first briefly introduce a commonly used framework for analyzing time-to-event outcomes, i.e., *the proportional hazards (PH) models*. Note that, for the sake of simplicity, we only focus on right-censored data without competing risks in this thesis.

This section first introduces the structure of time-to-event data, then explains key quantities in proportional hazards (PH) models. Next, it reviews two commonly used estimators and concludes with assumption checking using Schoenfeld residuals.

Data Structure

In time-to-event data analysis, the data structure is typically the following. For each individual i ($i = 1, \dots, n$), we represent the observed data as Equation 1.

$$\mathcal{D}_i = \{\mathbf{x}_i, T_i, \delta_i\}, \quad (1)$$

where:

- $\mathbf{x}_i \in \mathbb{R}^{1 \times p}$ is consisted of covariate vectors,
- T_i is the observed time to event or censoring,
- $\delta_i = \mathbb{I}(T_i \leq C_i)$ is the event indicator.

Specifically, $\delta_i = 1$ if the event is observed (i.e., $T_i^* = T_i$), and $\delta_i = 0$ if the observation is right-censored (i.e., $T_i^* > T_i$), where T_i^* denotes the true event time. C_i denotes the censoring time.

Hazard function and other relevant quantities

In PH models, a central quantity is the *hazard function*, which describes the instantaneous risk of an event occurring at time t for individual i , given that the event has not yet occurred. Formally, the hazard function is given by Equation 2, and its specification under the PH model is shown in Equation 3.

$$h_i(t) = \lim_{\Delta t \rightarrow 0} \frac{P(t \leq T_i^* < t + \Delta t \mid T_i^* > t)}{\Delta t} \quad (2)$$

$$= h_0(t) \exp(\boldsymbol{\eta}_i) \quad (3)$$

where $h_0(t)$ is the baseline hazard function and $\boldsymbol{\eta}_i$ represents the linear combination of covariates for individual i , i.e., $\mathbf{x}_i \boldsymbol{\beta}$. This formulation assumes (1) *non-informative censoring*, i.e., for each individual i , the censoring time C_i is independent of the true event time T_i^* given the covariates $\mathbf{x}_i$; and (2) *proportional hazard*, i.e., the effect of a covariate is constant over time.

In addition to the hazard function, several other quantities are also fundamental in PH models:

- **Cumulative hazard function:**

$$H_i(t) = \int_0^t h_i(u) du = H_0(t) \exp(\boldsymbol{\eta}_i),$$

which represents the accumulated risk up to time t for individual i .

- **Survival function:**

$$S_i(t) = \exp(-H_i(t)) = S_0(t)^{\exp(\boldsymbol{\eta}_i)},$$

giving the probability that individual i survives beyond time t .

- **Density function:**

$$f_i(t) = S_i(t) \cdot h_i(t),$$

which describes the probability distribution of the event time.

Estimators in PH models

Maximum Likelihood Estimator (MLE)

With the time-to-event data structure and the relevant quantities defined in the proportional hazards (PH) model, we now move on to one of the two primary estimators for the regression

coefficients β in the linear predictor η —namely, the *maximum likelihood estimator (MLE)*. For n independent subjects, the MLE is given in Equation 4.

$$\begin{aligned} \log L(\mathcal{D}_i \mid \beta, \text{other parameters}) = \sum_{i=1}^n \left\{ \mathbb{I}(\delta_i = 0) \log[S_i(t)] \right. \\ \left. + \mathbb{I}(\delta_i = 1) (\log[h_i(t)] + \log[S_i(t)]) \right\}, \end{aligned} \quad (4)$$

The indicator function $\mathbb{I}(\cdot)$ ensures that the MLE accounts appropriately for both censored and uncensored observations.

Note that as shown in Equation 4, defining the MLE involves specifying one of the following functions: $S_0(t)$, $H_0(t)$, or $h_0(t)$. In this thesis, we focus on specifying $h_0(t)$ by considering both parametric and flexible parametric approaches for the sake of simplicity and practicality. That is, we consider the following:

1. **(Parametric) Exponential:** assumes a constant baseline hazard function, where true event times T^* follow $\text{Exp}(\lambda)$.
2. **(Parametric) Weibull:** assumes a time-varying baseline hazard function, where true event times T^* follow $\text{Weibull}(\lambda, v)$.
3. **(Flexible Parametric) B-Splines-based:** approximates the log baseline hazard function using B-Splines, i.e., $\log h_0(t) \approx \sum_{l=1}^L \gamma_l B_l(t; k, \zeta)$, where k denotes the knot vector, ζ is the degree of the B-Splines basis, and γ_l are the coefficients associated with the B-Splines basis functions.

Based on these approaches, we can obtain other quantities as summarized in Table 1. Note that, in contrast to the closed-form expressions of the cumulative hazard function and survival function under the assumption of a constant or time-varying baseline hazard (i.e., when true event times T^* follow an Exponential or Weibull distribution), approximating the log-baseline hazard function using the B-splines method does not yield closed-form solutions for $H(t)$ or $S(t)$. Therefore, for the B-splines approach in this thesis, we approximate $H(t)$ from the baseline hazard function $h(t)$ using the Gauss–Kronrod quadrature method with 15 nodes (see Equation 5). This method is chosen because it provides a smooth and accurate approximation of the integral of the baseline hazard function (Brilleman et al., 2020).

$$\int_{u=0}^{T_i} h_i(u) du \approx \frac{T_i}{2} \sum_{q=1}^Q w_q h_i \left(\frac{T_i(1 + w_q)}{2} \right) \quad (5)$$

where w_q are the quadrature weights and $Q = 15$ is the number of nodes.

Table 1: PH models under different baseline hazard assumptions

Assumption for t	Exponential $h_0(t) = \lambda$	Weibull $h_0(t) = \lambda vt^{v-1}$	B-Splines Approximation $h_0(t) = \exp\left(\sum_{l=1}^L \gamma_l B_l(t; k, \zeta)\right)$
Parameter(s)	Scale $\lambda > 0$	Scale $\lambda > 0$ Shape $v > 0$	Coefficients γ_l for B-spline basis
$h(t)$	$\lambda \exp(\eta_i)$	$\lambda vt^{v-1} \exp(\eta_i)$	$\exp\left(\sum_{l=1}^L \gamma_l B_l(t; k, \zeta) + \eta_i\right)$
$H(t)$	$\lambda \exp(\eta_i)t$	$\lambda \exp(\eta_i)t^v$	$\int_0^t h(u) du$
$S(t)$	$\exp(-\lambda \exp(\eta_i)t)$	$\exp(-\lambda \exp(\eta_i)t^v)$	$\exp(-H(t))$

Note: $B_l(t; k, \zeta)$ in Table 1 denote the l^{th} basis function ($l = 1, \dots, L$) of a B-Spline of degree ζ , evaluated at the knot vector $k = \{k_1, \dots, k_J\}$. Each basis term has an associated coefficient γ_l , which is estimated from the data. In this thesis, we employ the `bSpline` function from the `splines2` R package (Wang & Yan, 2021) to construct cubic B-Splines (i.e., $\zeta = 3$) for modeling the log-baseline hazard as a smooth function of time. The B-Spline basis excludes an intercept. The two boundary knots are positioned at 0 and the maximum of observing times. Thus, the estimated baseline hazard function can span the entire follow-up period, similar to the coverage of the classic Kaplan-Meier estimator. Additionally, two internal knots are placed at evenly spaced quantiles of the observed event times, which concentrates knots in regions with higher event density.

Maximum Partial Likelihood Estimator (MPLE)

Besides the MLE, another important estimator of β in the linear combination η is the *maximum partial likelihood estimator (MPLE)*, which has been introduced by (Cox, 1975) as Equation 6. Afterwards, this work has been updated by (Efron, 1977) to account for tied event times, i.e., multiple events occur at the same time (see Equation 7). Compared to the MLE in Equation 4, MPLE in Equation 6 and 7 can estimated β without specifying the baseline hazard $h_0(t)$ and thus become more computationally efficient and applicable than MLE.

$$\log PL(\mathcal{D}_i \mid \beta) = \sum_{i=1}^n \delta_i \left[\mathbf{x}_i \beta - \log \left(\sum_{j \in R_i} \exp(\mathbf{x}_j \beta) \right) \right] \quad (6)$$

where R_i is the risk set when i -th event occurs.

$$\log PL_{\text{Efron}}(\beta) = \sum_{j=1}^J \left[\sum_{i \in D_j} \mathbf{x}_i \beta - \sum_{r=0}^{d_j-1} \log \left(\sum_{k \in R_j} \exp(\mathbf{x}_k \beta) - \frac{r}{d_j} \sum_{i \in D_j} \exp(\mathbf{x}_i \beta) \right) \right] \quad (7)$$

where J is the number of distinct event times, d_j is the number of events at the j -th distinct event time, D_j is the set of tied events at the j -th distinct event time, and R_j is the risk set at the j -th distinct event time.

Assumption checking

As discussed earlier, the Proportional Hazards (PH) models and their associated functions/quantities rely on two key assumptions: non-informative censoring and proportional hazard. Non-informative censoring typically depends on how the data is recorded. In contrast, proportional hazard needs to be properly checked based on the output of the model. One common approach to assess the adequacy of the proportional hazards assumption for both categorical and continuous covariates is to plot the Schoenfeld residuals against the event times.

The Schoenfeld residual is a partial residual, defined as the vector at i -th event time,

$$\hat{r}_i = (\hat{r}_{i1}, \hat{r}_{i2}, \dots, \hat{r}_{ip})^T$$

where

$$\hat{r}_{ij} = x_{ij} - \hat{E}(x_{ij} | R_i),$$

with

$$\hat{E}(x_{ij} | R_i) = \frac{\sum_{k \in R_i} x_{kj} \exp(\mathbf{x}_k \hat{\boldsymbol{\beta}})}{\sum_{k \in R_i} \exp(\mathbf{x}_k \hat{\boldsymbol{\beta}})}, \quad (8)$$

where R_i denotes the set of individuals at risk at i -th event time (i.e., under observation when the i -th event occurs; see Schoenfeld, 1982), and $\hat{\boldsymbol{\beta}}$ is the estimated coefficient vector.

In the presence of tied event times, Equation 8 will be updated according to Equation 7, ending up with Equation 9.

$$\hat{E}_{ij}^{\text{Efron}} = \frac{1}{d_i} \sum_{r=1}^{d_i} \frac{\sum_{k \in R_i} x_{kj} e^{\mathbf{x}_k \hat{\boldsymbol{\beta}}} - \frac{r-1}{d_i} \sum_{k \in D_i} x_{kj} e^{\mathbf{x}_k \hat{\boldsymbol{\beta}}}}{\sum_{k \in R_i} e^{\mathbf{x}_k \hat{\boldsymbol{\beta}}} - \frac{r-1}{d_i} \sum_{k \in D_i} e^{\mathbf{x}_k \hat{\boldsymbol{\beta}}}} \quad (9)$$

where R_i is the risk set at i -th distinct event time; D_i is the set of tied events at i -th distinct event time; $d_i = |D_i|$ is the number of tied events at i -th distinct event time; and $\hat{\boldsymbol{\beta}}$ is the estimated coefficient vector.

If the proportional hazards assumption stands, the Schoenfeld residuals have a sample path of a random walk, i.e., their values fluctuate randomly over time without exhibiting systematic trends. Therefore, they are useful in assessing time trend or lack of proportionality.

Related work

Building on the background of proportional hazards (PH) models and their estimators, we now review existing methods for addressing the problem mentioned in the Introduction when detecting interaction effects in survival or regression settings. These methods can be grouped into

frequentist and Bayesian approaches. One prominent frequentist approach in the regression setting is the *two-stage method*. The two-stage method begins with a model containing only main effects, and interaction terms are then added only if their corresponding main effects are statistically significant (Hao & Zhang, 2017). While this approach helps mitigate the quadratic growth of interaction terms as the number of main terms increases, it may be less applicable for small datasets due to concerns about double-dipping, i.e., using the same data to identify significant main effects and then to test interactions. Alternative models, such as lasso regression with penalties for interaction terms, often struggle to accommodate large differences in parameter strengths between main effects and interactions. Similarly, adaptive lasso, which weights penalties by the reciprocal of absolute OLS coefficients, is limited by the stability of the OLS estimates.

In contrast, Bayesian approaches offer more integrated solutions, such as the *linked shrinkage prior*. Specifically, as shown in Equation 10, in the context of linear regression, van de Wiel et al. (2024) have linked main and interaction effects through the product $\tau_j \tau_k$ in the prior for β_{jk} . This setting enables symmetric treatment of strong and weak main effects. Each τ_j is specified as $\mathcal{C}^+(0, 1)$, which has been reported to exhibit strong shrinkage properties (Gelman et al., 2008). Additionally, a shared shrinkage parameter τ_{int} is introduced for all interaction terms, following $\mathcal{U}(0.01, 1)$. This shared parameter reflects the common belief that interaction effects are, on average, weaker than main effects, while its lower bound prevents complete shrinkage to zero, which would be undesirable in sparse settings. For categorical covariates, van de Wiel et al. (2024) modify the summation over j, k to exclude interactions between different levels of the same factor. In other words, they consider only interaction effects between distinct factors, explicitly excluding interactions among mutually exclusive levels within a single categorical variable. Together, the linked shrinkage prior demonstrates clear advantages in detecting interaction effects in regression models. Note that other priors (e.g., the horseshoe prior), originally developed for high-dimensional data analysis rather than detecting interaction effects, can also be applied in this context.

$$\begin{aligned}
Y_i &= \alpha + \sum_{j=1}^p \beta_j x_{ij} + \sum_{j < k} \beta_{jk} x_{ij} x_{ik} + \varepsilon_i \\
\alpha &\sim \mathcal{N}(0, 10^2) \\
\beta_j &\sim \mathcal{N}(0, \sigma^2 \tau_j) \\
\beta_{jk} &\sim \mathcal{N}(0, \sigma^2 \tau_j \tau_k \tau_{\text{int}}) \\
\varepsilon_i &\sim \mathcal{N}(0, \sigma^2) \\
\tau_j &\sim \mathcal{C}^+(0, 1) \\
\tau_{\text{int}} &\sim \mathcal{U}(0.01, 1) \\
\sigma^2 &\sim \text{IG}(1, 0.001)
\end{aligned} \tag{10}$$

where $\mathcal{D} = \{(Y_i, \mathbf{x}_i)\}_{i=1}^n$ is n independent observations; Y_i is the response variable, and $\mathbf{x}_i$ is the

covariate vector; x_{ij} is the j th covariate for individual i , with $j = 1, \dots, p$.

The Current Study

In this study, we apply the linked shrinkage prior to Bayesian proportional hazard (PH) models to enable accurate estimation and selection of interaction effects in explaining graft loss in kidney transplantation. To this end, we employ both the maximum likelihood estimator (MLE) in Equation 4 and the maximum partial likelihood estimator (MPLE) in Equation 7. When employing MLE, for priors of the coefficients β in η , we adopt the prior structure from Equation 10 by replacing the response variable Y_i with the linear combination of covariates η . For ancillary parameters that employed to model $h_0(t)$, in this thesis, we specify their prior distributions as shown in Table 2.

Table 2: Prior distributions for ancillary parameters under different baseline hazard specifications

Parameter	Exponential	Weibull	B-Splines Approximation
Scale parameter λ	Lognormal(0, 100)	Lognormal(0, 100)	—
Shape parameter v	—	Lognormal(0, 100)	—
B-spline coefficients γ_l	—	—	$\mathcal{N}(0, 100)$

Regarding MPLE, as mentioned earlier, MPLE in Equations 6 and 7 effectively removes the nuisance baseline hazard function. Although this simplification improves computational efficiency, it poses challenges for classical Bayesian methods. Specifically, the posterior distribution is derived from the full likelihood estimator and the prior distribution via $P(\theta | \text{Data}) \propto L(\text{Data} | \theta) P(\theta)$. Nonetheless, Kalbfleisch (1978) demonstrated that using MPLE in Bayesian PH models approximates the marginal posterior distribution of β under the assumption that $H_0(t)$ follows a gamma process prior. This result was later extended by Kim and Kim (2009) to accommodate tied event times. Together, these findings provide strong theoretical justification for applying the linked shrinkage prior to Bayesian PH models by incorporating MPLE. Therefore, we employ MPLE in this study and define the priors for β in the same way as in the MLE-based application.

Validating reformulated method on simulated study

Simulated data generation

To validate the performance of our Bayesian proportional hazards (PH) models that incorporate a linked shrinkage prior, we first conduct a simulation study using data generated via the `simsurv`

package (Brilleman et al., 2021). The `simsurv` package allows flexible specification of both the baseline hazard function $h_0(t)$ and the linear combination of covariates $\boldsymbol{\eta}$. In this thesis, we set the ground-truth baseline hazard function to increase over time, where true event times follow a Weibull distribution with parameters $\lambda = 2$ and $v = 10$. To reflect realistic conditions, we set the event occurrence rate to 75% and adjust the censoring accordingly. For the linear combination of covariates $\boldsymbol{\eta}$, we generate $p = 10$ moderately collinear continuous covariates for $n = 200$ samples, resulting in $q = \frac{p(p-1)}{2} = 45$ two-way interaction terms.

We account for three possible scenarios of interaction terms regarding the main effects: (1) interaction effects between two non-zero main effects; (2) interaction effects between one non-zero and one zero main effect; and (3) interaction effects between two zero main effects. To this end, five main effects are assigned non-zero values (ranging from small to large), while the remaining five are set to zero. A separate testing dataset of $n' = 100$ samples is generated using the same configuration for the sake of prediction. The detailed information about the values of $\boldsymbol{\beta}$ in the simulation setting is as follows:

- **Sample size:** $n = 200$ (estimation/training), $n' = 100$ (testing)
- **Covariates:** $p = 10$ continuous variables
- **Interactions:** $q = \frac{p(p-1)}{2} = 45$ two-way interactions
- **Main effects:** $\beta_1 = 0.5$, $\beta_2 = -0.4$, $\beta_3 = 0.3$, $\beta_4 = 0.2$, $\beta_5 = 0.1$; β_6 to $\beta_{10} = 0$
- **Non-zero interactions:**
 - Between two non-zero main effects: $\beta_{12} = 0.3$, $\beta_{13} = 0.2$, $\beta_{15} = -0.1$, $\beta_{23} = -0.1$, $\beta_{24} = 0.3$, $\beta_{34} = 0.2$
 - Between one non-zero and one zero main effect: $\beta_{16} = 0.3$, $\beta_{19} = 0.2$, $\beta_{36} = -0.2$, $\beta_{47} = 0.1$
 - Between two zero main effects: $\beta_{67} = 0.3$, $\beta_{68} = 0.1$, $\beta_{9,10} = -0.2$
- **Remaining interactions:** The other $45 - 12 = 33$ interaction terms are set to zero.
- **Covariate generation:** Each covariate vector $\mathbf{x}_i = (x_{i1}, \dots, x_{i10})$ is drawn from a multi-variate normal distribution: $\mathbf{x}_i \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma})$, where $\Sigma_{jj} = 1$ and $\Sigma_{jk} = 0.3$ for $j \neq k$.

Performance metrics

With the simulated data generated as described previously, we evaluate our application of linked shrinkage prior to Bayesian proportional hazards (PH) models with the following metrics:

1. Parameter estimation:

- *Bias:*

$$\text{Bias} = \frac{1}{B} \sum_{b=1}^B \left(\hat{\beta}^{(b)} - \beta \right)$$

- *Variance:*

$$\text{Variance} = \frac{1}{B} \sum_{b=1}^B \left(\hat{\beta}^{(b)} - \bar{\hat{\beta}} \right)^2, \quad \bar{\hat{\beta}} = \frac{1}{B} \sum_{b=1}^B \hat{\beta}^{(b)}$$

- *Root Mean Squared Error (rMSE):*

$$\text{rMSE} = \sqrt{\frac{1}{B} \sum_{b=1}^B \left(\hat{\beta}^{(b)} - \beta \right)^2}$$

Where B denotes the number of such simulation procedures in the study; $\hat{\beta}^b$ represents the median value of the corresponding posterior distribution in the b -th simulation; and β is the ground-truth value specified in the simulation. Note that, due to the bias-variance trade-off, a well-performing model should aim to minimize rMSE rather than bias and variance simultaneously. In this study, we report bias and variance to decompose rMSE.

2. Variable selection:

- *Type I Error/False Discovery Rate (FDR):*

$$\text{FDR} = \frac{\text{FP}}{\text{FP} + \text{TN}}$$

- *Power (1 - Type II Error):*

$$\text{Type II Error} = \frac{\text{FN}}{\text{FN} + \text{TP}}$$

Where FP, TN, FN, and TP are defined in Table 3. The parameter inference is based on its corresponding 95% credible interval.

Table 3: Confusion Matrix

	Actually Positive	Actually Negative
Predicted Positive	True Positives (TP)	False Positives (FP)
Predicted Negative	False Negatives (FN)	True Negatives (TN)

3. Prediction:

- *Concordance Index (C-index)*: Measures the model’s ability to correctly rank survival times. It is defined as the proportion of all usable patient pairs for which the predicted and observed outcomes are concordant. A higher C-index indicates better discriminative performance.

Note that the prediction metric, i.e., C-index, presented in this study are not the primary focus. They are included for two main reasons: (1) to serve as a complementary elements that support the overall analysis; (2) to evaluate of model performance in the context of real-world data.

Results of method performance: different ways of specifying $h_0(t)$

Using simulated data and the evaluation metrics described earlier, we first examine our Bayesian proportional hazard (PH) models that incorporate linked shrinkage prior under different specifications of $h_0(t)$. Specifically, as mentioned previously, we consider full Bayesian parametric models based on MPLE, where we (1) assume a constant baseline hazard function, where true event times T^* follow $\text{Exp}(\lambda)$; (2) assume a time-varying baseline hazard function, where true event times T^* follow $\text{Weibull}(\lambda, v)$; and (3) approximate $\log h_0(t)$ using B-Splines. Next, we consider partial Bayesian model based on MPLE (from now on **PartialLikelihood**). Since the simulated true event times T^* follow a Weibull distribution, we expect **Weibull** to achieve the best performance across all evaluation criteria (Lázaro et al., 2021). Conversely, **Exponential** is anticipated to perform the worst due to its misspecified distributional assumption.

As a result (see Figure 1), **Weibull** indeed demonstrates the best performance in estimating β , selecting relevant terms, and achieving high prediction accuracy. In contrast, **Exponential** exhibits the worst performance across these metrics. **B-Splines** and **PartialLikelihood** fall between these two extremes. We also visualize one of the estimated baseline hazard functions obtained from **Exponential**, **Weibull** and **B-Splines** (see Figure 2) and observed a similar result. Note that the good performance of **Weibull**, **B-Splines**, and **PartialLikelihood** comes at the cost of slightly higher variance for $\hat{\beta}$ compared to **Exponential**. This is reasonable given their much smaller bias and lower rmSE relative to **Exponential**. Together, our Bayesian proportional hazard (PH) models that incorporate linked shrinkage prior are correctly specified. However, given that, in practice, the true baseline hazard function remains commonly unknown, in the following sections we only focus on **B-Splines** and **PartialLikelihood**.

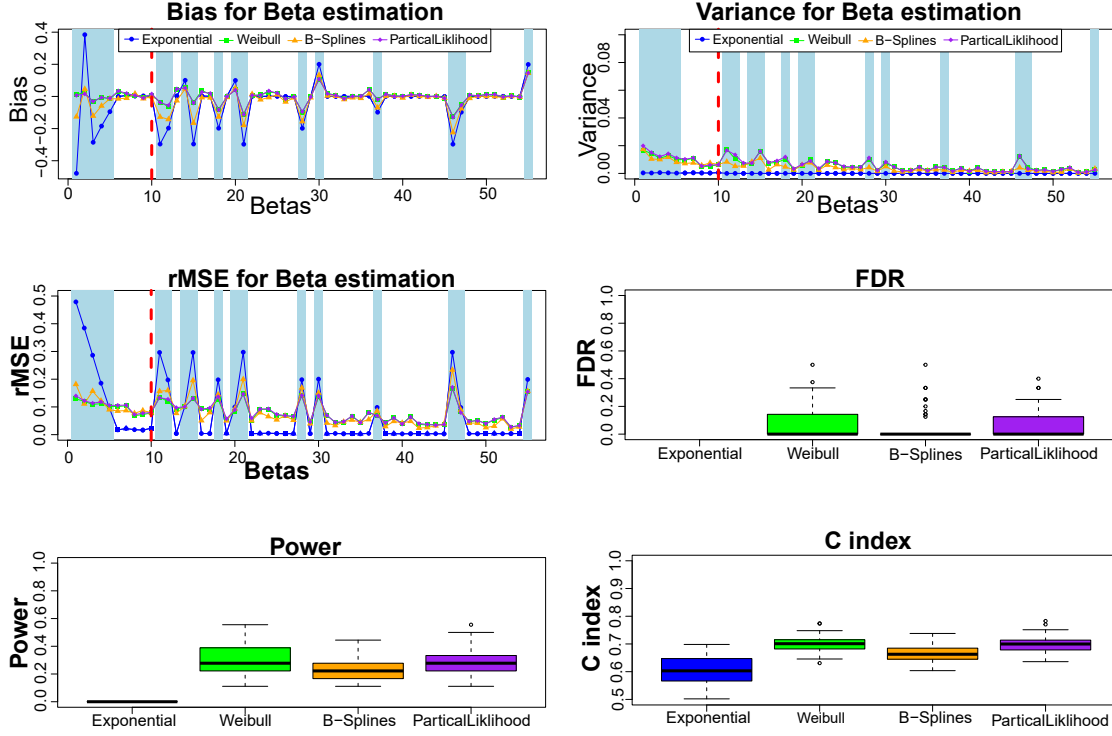


Figure 1: Performance of the applied linked shrinkage prior to Bayesian PH models across different specifications of $h_0(t)$. To the left of the red dashed vertical line are the main effects, while to the right are the interaction effects. **Weibull** specification demonstrates the best performance in estimating β , selecting relevant terms, and achieving high prediction accuracy. **Exponential** specification exhibits the weakest performance across these metrics, whereas **B-Splines** and **PartialLikelihood** fall between these two extremes.

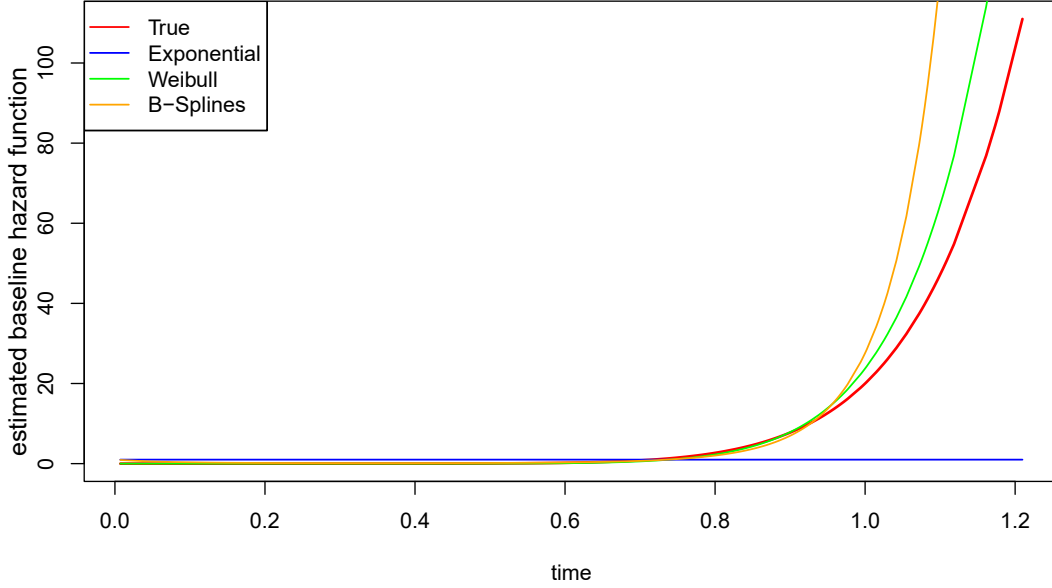


Figure 2: An example of $\hat{h}_0(t)$ from different methods. **Weibull** demonstrates the best performance in estimating $h_0(t)$. **Exponential** exhibits the weakest performance in estimating $h_0(t)$. **B-Splines** falls between these two extremes.

Results of method performance: linked shrinkage vs. non-informative prior

With **B-Splines** and **PartialLikelihood**, we now move on to validating the advantage of the linked shrinkage prior in detecting interaction effects using the simulated data. To this end, we employ the corresponding models that incorporates non-informative prior as a benchmark. For the non-informative prior, we simply set all prior distributions for β to $\mathcal{N}(0, 100)$, including those for both main effects and interaction effects. From now on, the corresponding models are **B-Splines_no** and **PartialLikelihood_no**. Note that the estimated coefficients $\hat{\beta}$ obtained from Bayesian PH models that incorporates this non-informative prior setting is equivalent to those derived from the classical cox regression from a frequentist perspective.

As shown in Figure 3, the **B-Splines** and **PartialLikelihood** consistently outperform **B-Splines_no** and **PartialLikelihood_no**. Again, no consistent performance difference is observed between **B-Splines** and **PartialLikelihood**. As an example, we also visualize an estimated baseline hazard function $\hat{h}_0$ obtained from **B-Splines** and **B-Splines_no**. Accordingly, **B-Splines** performs better than **B-Splines_no** in estimating baseline hazard function. Together, **B-Splines** and **PartialLikelihood** outperform their counterparts that employ non-informative priors in detecting interaction effects.

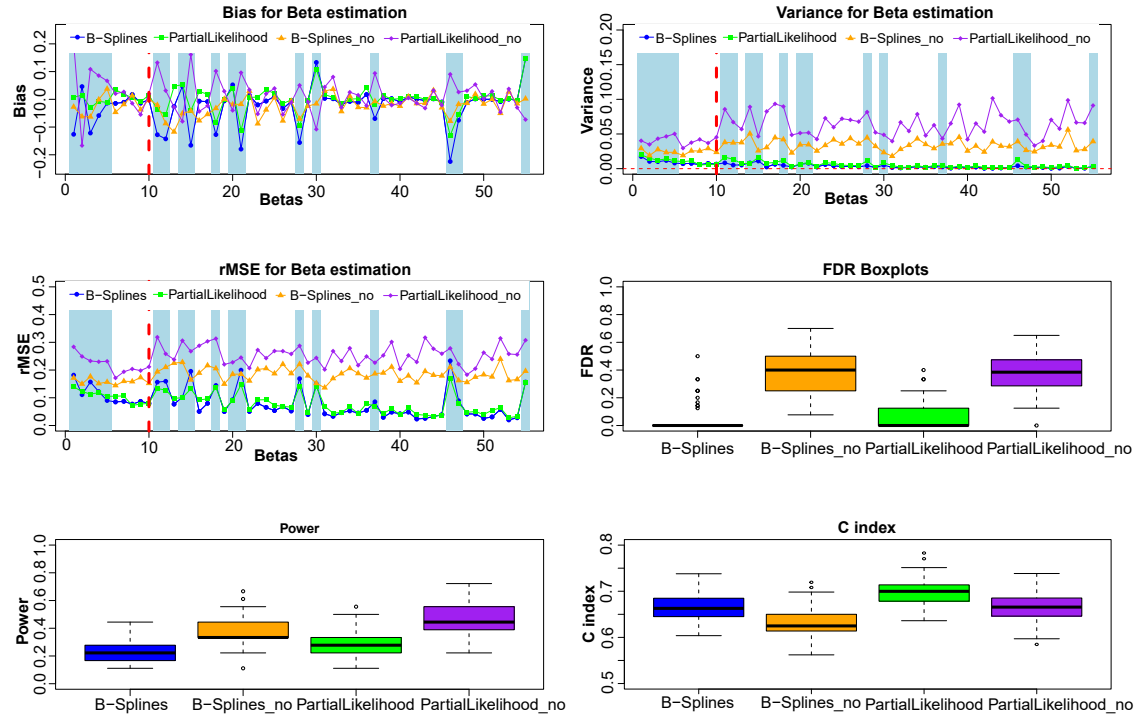


Figure 3: Performance of the Bayesian PH models incorporating the linked shrinkage prior is compared to those using non-informative priors. To the left of the red dashed vertical line are the main effects, while to the right are the interaction effects. **B-Splines** and **PartialLikelihood** consistently outperform their counterparts with non-informative priors across all evaluation metrics. No consistent difference is observed between **B-Splines** and **PartialLikelihood**.

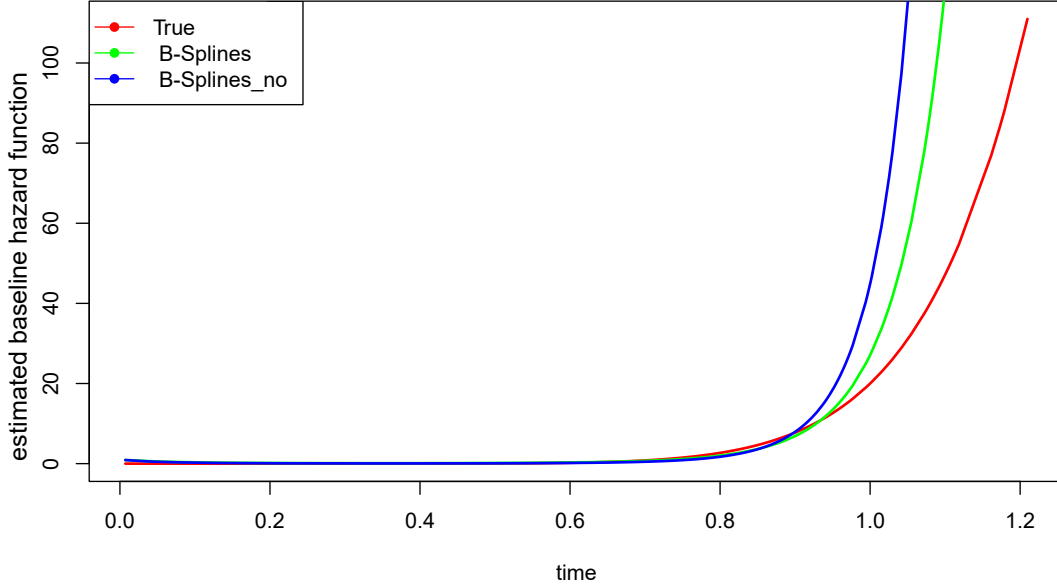


Figure 4: An example of $\hat{h}_0(t)$ from **B-Splines** and **B-Splines_no**. **B-Splines** performs better than **B-Splines_no** in estimating baseline hazard function.

Detecting interaction effects in kidney transplantation

With the validated **B-Splines** and **PartialLikelihood** approaches, we now proceed with the following: (1) apply **B-Splines** and **PartialLikelihood** to real-world data; and (2) focus on the original aim of this study, namely, accurate estimation and selection of interaction effects to better explain graft loss in kidney transplantation. To this end, we also include **B-Splines_no** and **PartialLikelihood_no** as benchmarks.

Regarding the data, we use kidney transplantation records from the national registry of the Dutch National Transplant Foundation (NOTR), which is the same dataset employed in Steenvoorden, Evers, et al. (2024) and Steenvoorden, Vogt, et al. (2024).

Data

With the kidney transplantation records obtained from NOTR, we first follow the same preprocessing procedure as described in Steenvoorden, Vogt, et al. (2024). After preprocessing, the resulting dataset (hereafter referred to as **complexData**) includes (1) 3008 observations, with 310 recorded events of graft loss; and (2) 14 covariates, comprising 8 continuous and 6 cate-

gorical variables. The continuous covariates are: `recipientage`, `dialysesvintage`, `donorage`, `lastcreatininemgdl`, `cold_ischaemic_period`, `HLA_mismatch`, `donorBMI`, and `recipient_BMI`. The categorical covariates are: `recipientsex` (Female/Male), `donorsex` (Female/Male), `smoking` (Yes/No), `typecadaveric` (DCD/DBD), `initialPrimaryDiseaseET` (Diabetes/Others), and `donorcauseofdeath` (Cardiac/Others).

We also consult domain experts to exclude certain covariates based on expert knowledge, resulting in a simplified dataset (hereafter referred to as `simpleData`). The excluded covariates are: `smoking` (Yes/No), `donorcauseofdeath` (Cardiac/Others), `dialysesvintage`, `recipientsex`, and `recipient_BMI`. The `simpleData` is included to ensure good convergence for PH models that incorporate non-informative priors.

Additionally, we introduce extra noise variables (i.e., `noise1` and `noise2`) into both `complexData` and `simpleData` to help detect false positives. Consequently, the final number of covariates for `complexData` is $p = 14 + 2 = 16$, while for `simpleData` it is $p = 9 + 2 = 11$. We consider all two-way interactions among covariates, resulting in $q = \binom{16}{2} = 120$ interaction parameters for `complexData` and $q = \binom{11}{2} = 55$ for `simpleData`. In total, this yields 136 terms for `complexData` and 66 terms for `simpleData`.

To prevent zero or extremely early event times, the observation time T_i in Equation 1 is shifted slightly to the right by adding 0.5. In this analysis, graft loss is treated as the event of interest, while death is considered a censored observation rather than a competing risk. In other words, we assume that death is unrelated to the risk of graft loss. Under this assumption, the model simply removes the patient from the risk set without introducing bias.

Descriptive data analysis for real-world data

Before analyzing the real-world dataset, we first visualize Kaplan–Meier curve of the kidney transplantation data (see Figure 5) and calculate the median follow-up time to evaluate the maturity and reliability of the follow-up for graft loss via the reverse Kaplan–Meier method (see Figure 6). Accordingly, there is a sharp decline in survival probability at the beginning, and the median follow-up time is 1437 days (approximately 3.9 years). This indicates that at least 50% of the patients were followed for this duration.

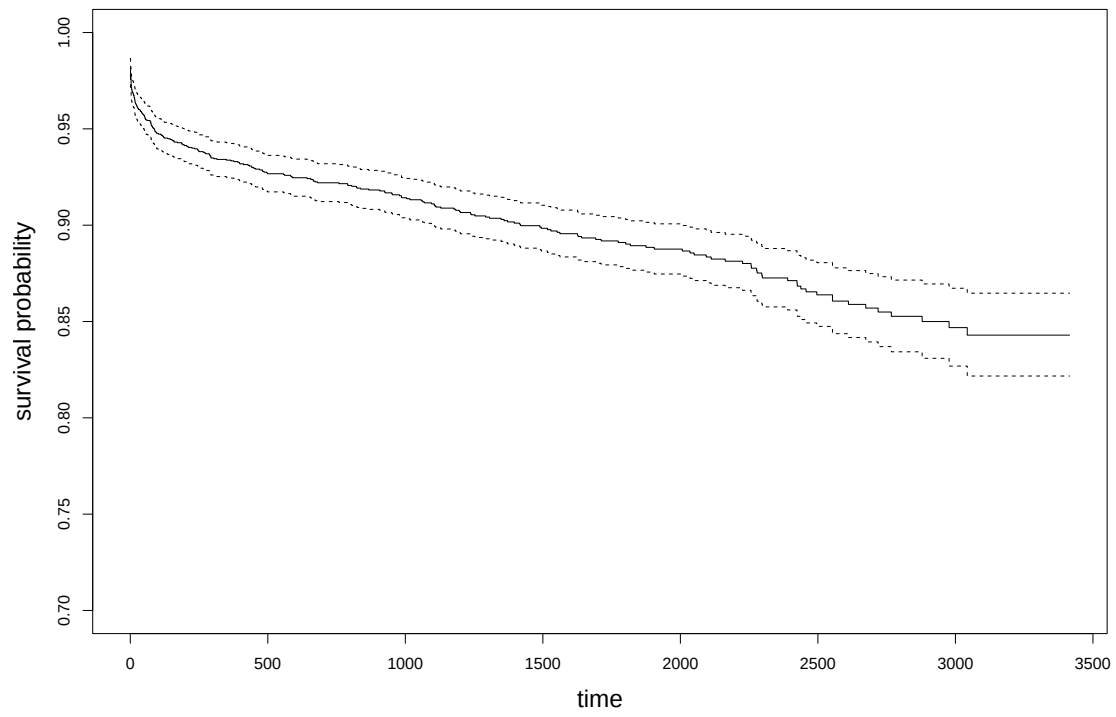


Figure 5: Kaplan–Meier curve of the real-world kidney transplantation dataset. There is a sharp decline in survival probability at the beginning.

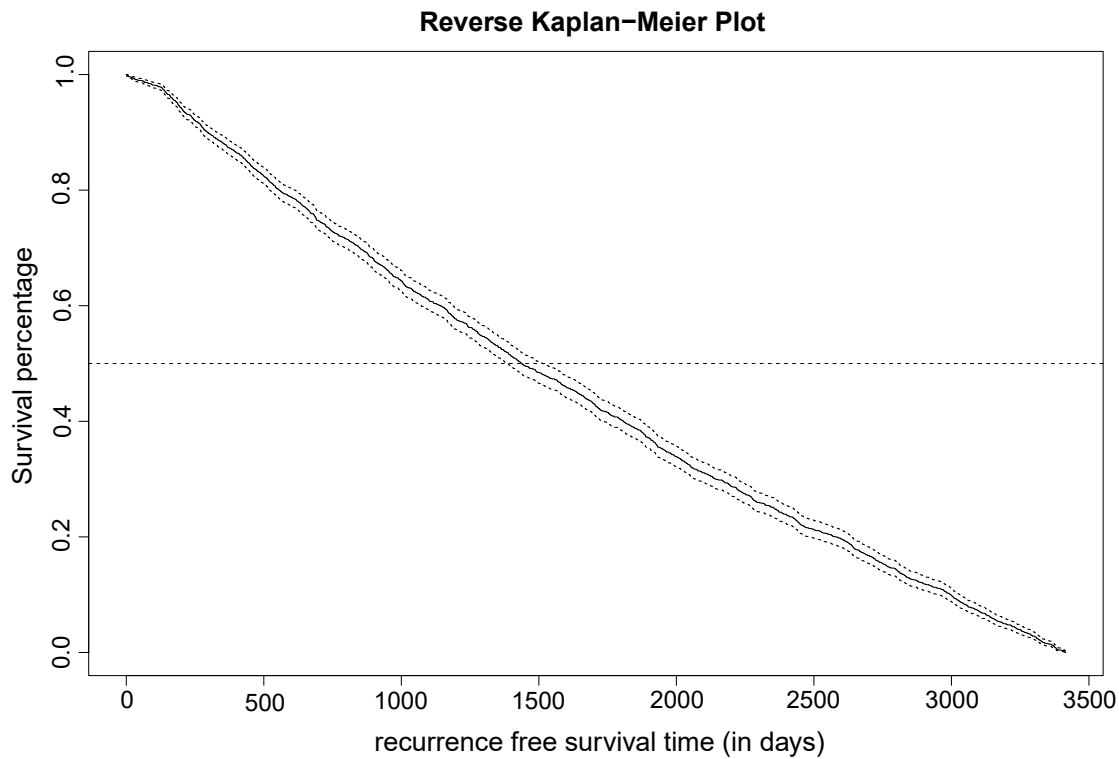


Figure 6: The reverse Kaplan–Meier curve for the real-world kidney transplantation dataset treats censoring times, including death, as “events”, while actual events (e.g., graft loss) are treated as censored. The estimated median follow-up time is 1,437 days (approximately 3.9 years).

We also assess collinearity among the covariates by analyzing pairwise correlations and evaluating the occurrence of each category in categorical covariates, separately for graft loss and censored events. As a result, we did not detect any strong collinearity, as shown in Figure 7. The occurrence of each category for graft loss and censored events is also within a reasonable range (see Table 4).

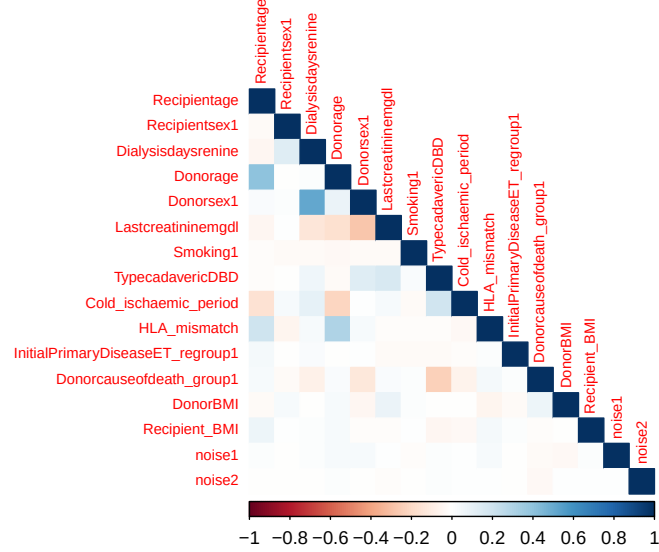


Figure 7: Pair-wise correlations among covariates in the dataset. No strong collinearity has been detected.

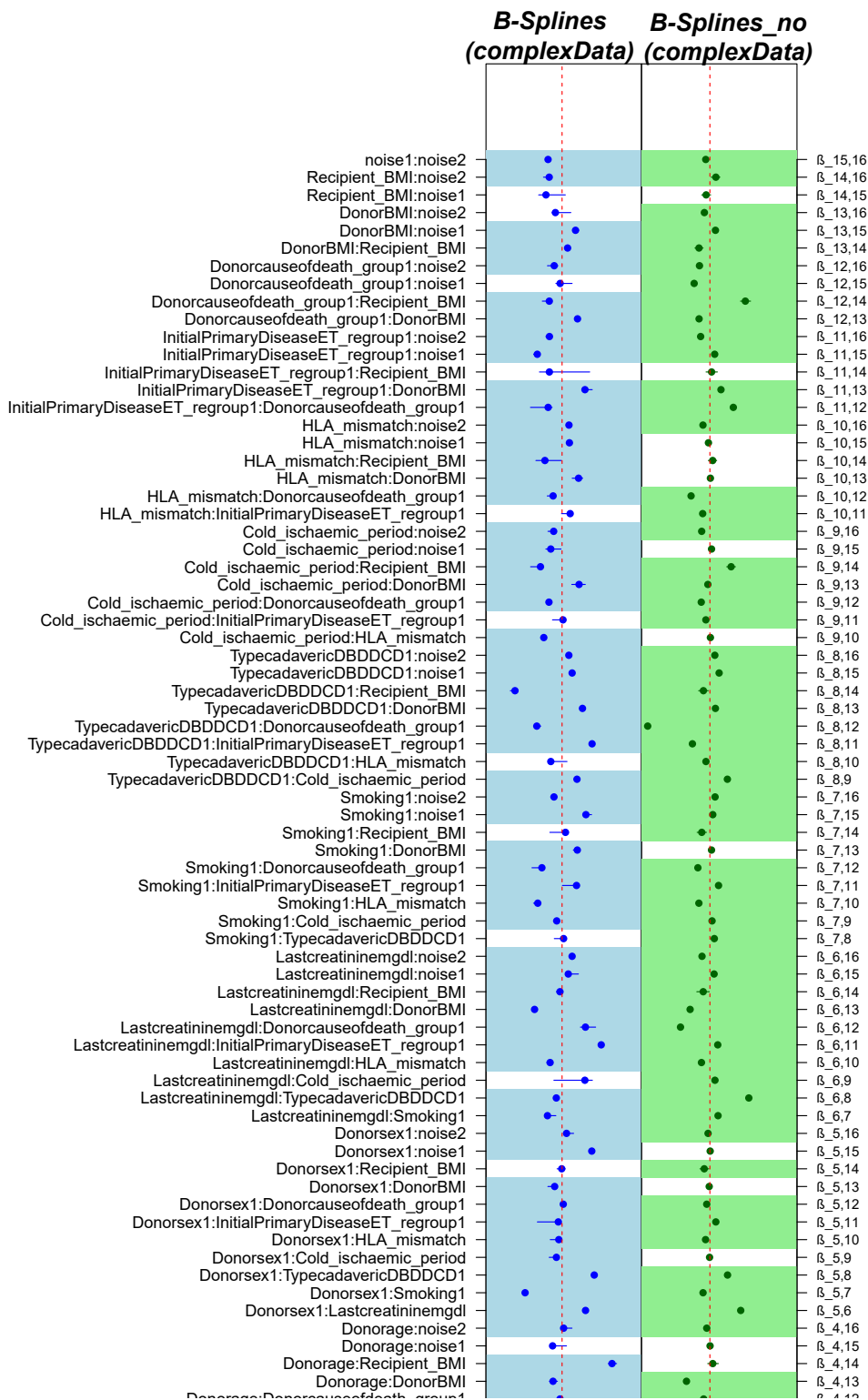
Table 4: Summary of the occurrence of each category in categorical covariates, separately for graft loss and censored events.

	Recipient sex		Donor sex		Smoking		Typecadaveric		Initial Primary Disease		Donor cause of death	
	Female	Male	Female	Male	Yes	No	DCD	DBD	Diabetes	Others	Cardiac	Others
Graft loss	124	186	161	149	184	149	175	135	22	288	54	256
Censored	979	1719	1193	1505	1575	1123	1619	1079	223	2475	487	2211

Results of kidney transplantation data

As we mentioned before, in this section, our methodological aim is to apply **B-Splines** and **PartialLikelihood** to real-world kidney transplantation data. To this end, first, as benchmark models, we include **B-Splines_{no}** and **PartialLikelihood_{no}**. Second, we generate some noise variables and include them in the kidney transplantation dataset. Last, we employ both **B-Splines** and **PartialLikelihood** to analyze **complexData** and **simpleData**.

As a result, for `complexData`, `B-Splines` and `B-Splines_no` detect an unrealistically large number of covariates (see Figure 8), including interactions between `noise1`, `noise2`, and other covariates. This is also the case when applying `B-Splines` and `B-Splines_no` to `simpleData` (see Figure 9). Taken together, these findings suggest that both `B-Splines` and `B-Splines_no` are not reliable when applied to kidney transplantation data. Therefore, in the remainder of this section, we focus on `PartialLikelihood`.



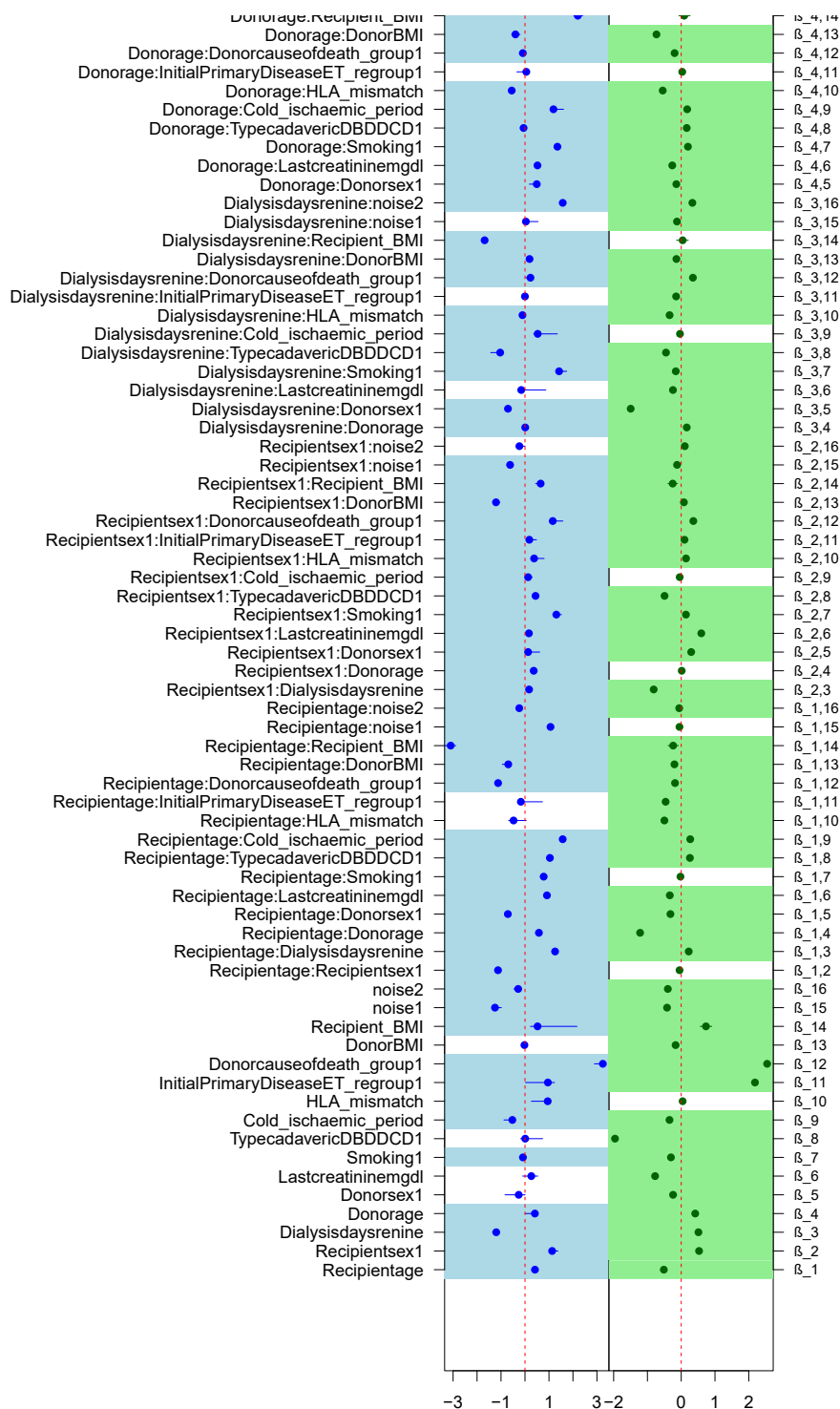


Figure 8: Results for `complexData` using `B-Splines` and `B-Splines_no`. Horizontal bars indicate the corresponding 95% credible intervals. Shaded terms denote the effects selected by each method. Both `B-Splines` and `B-Splines_no` yield similar outcomes, identifying unrealistically large number of covariates, including interactions between `noise1`, `noise2`, and other variables. Together, these methods appear unreliable for analyzing `complexData`.

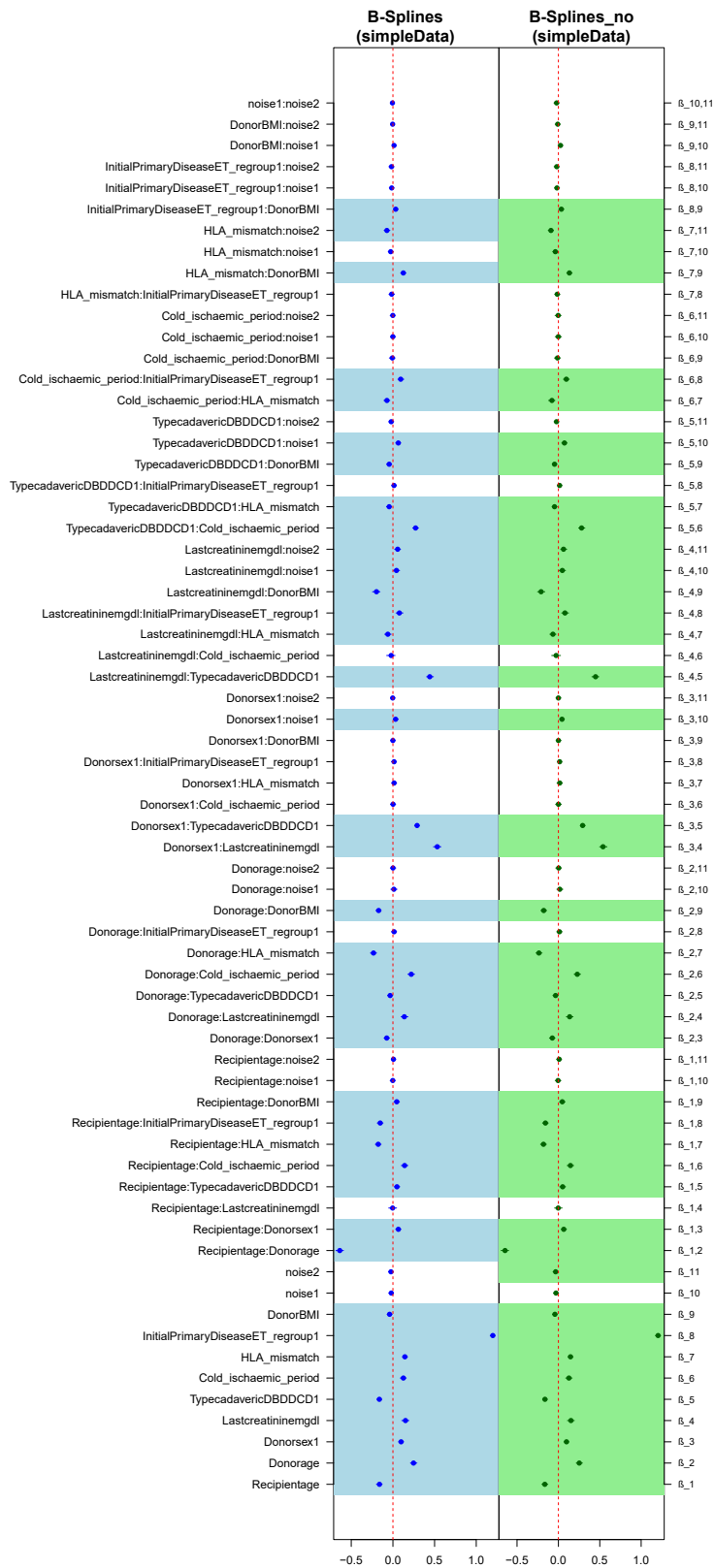
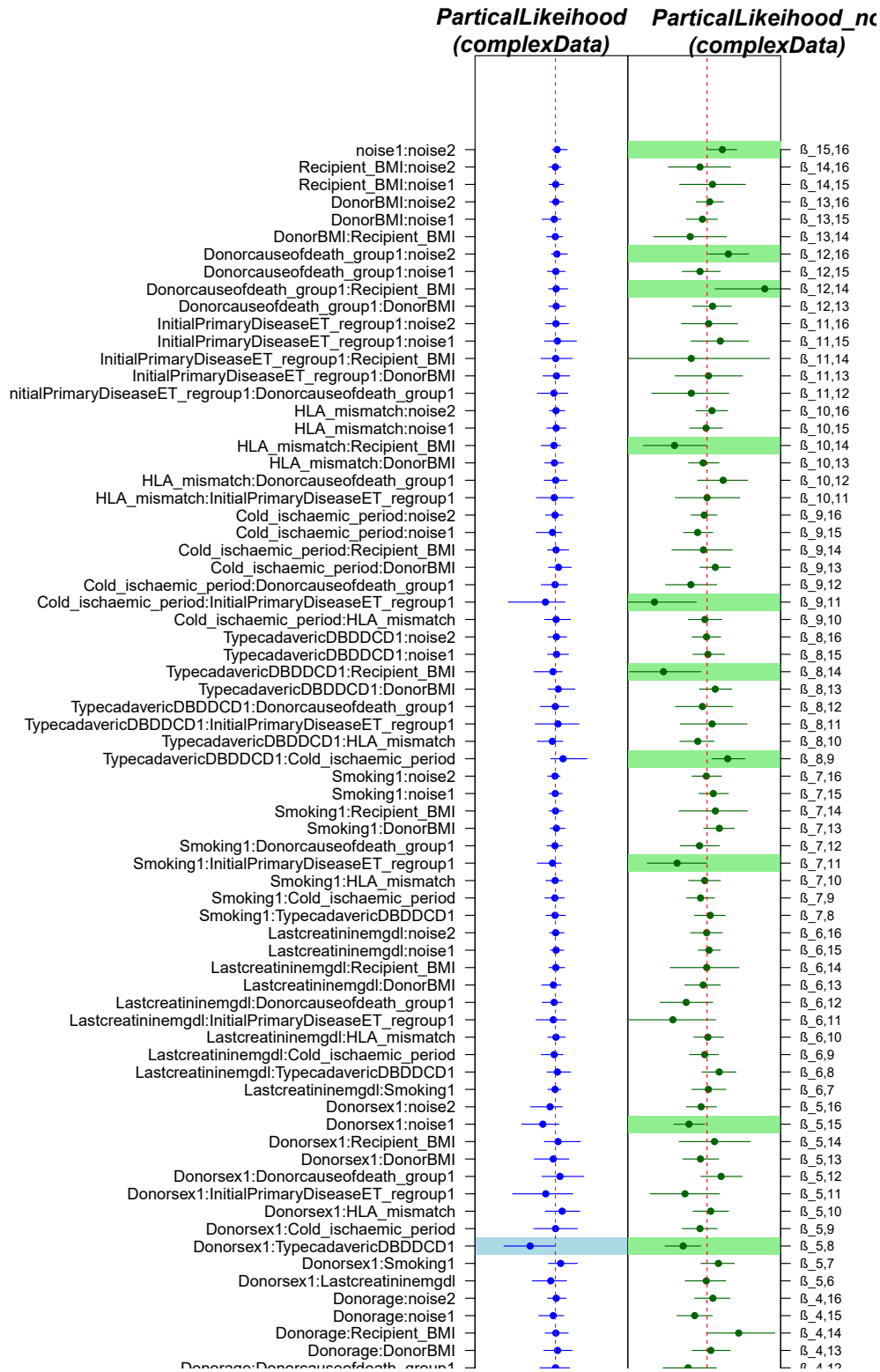


Figure 9: Results for `simpleData` using B-Splines and B-Splines_{no}. Horizontal bars represent the 95% credible intervals, and shadowed terms indicate the effects selected by each method. Both approaches detect unrealistically numerous covariates, including the main effects of `noise1` and `noise2` as well as their interactions with other covariates. Therefore, these methods appear unreliable for analyzing `simpleData`.

Regarding `PartialLikelihood`, its non-informative prior counterpart `PartialLikelihood_no` also detects interaction effects with noise variables (see Figure 10 and 11). This is the case when applying `PartialLikelihood_no` to both `complexData` and `simpleData`. In contrast, `PartialLikelihood` does not detect any effects related to the noise variables. More concretely, when applying `PartialLikelihood` to `complexData`, the main effects `dialysesvintage` and `Donorage`, as well as the interaction effects `dialysesvintage:Donorsex1` and `Donorsex1:Typecadaveric1`, are detected. When applying `PartialLikelihood` to `simpleData`, the main effects `Donorage` and `HLA_mismatch` are detected. Note that the interaction effect `Donorsex:TypecadavericDBDDCD` has been previously reported by Steenvoorden, Vogt, et al. (2024).



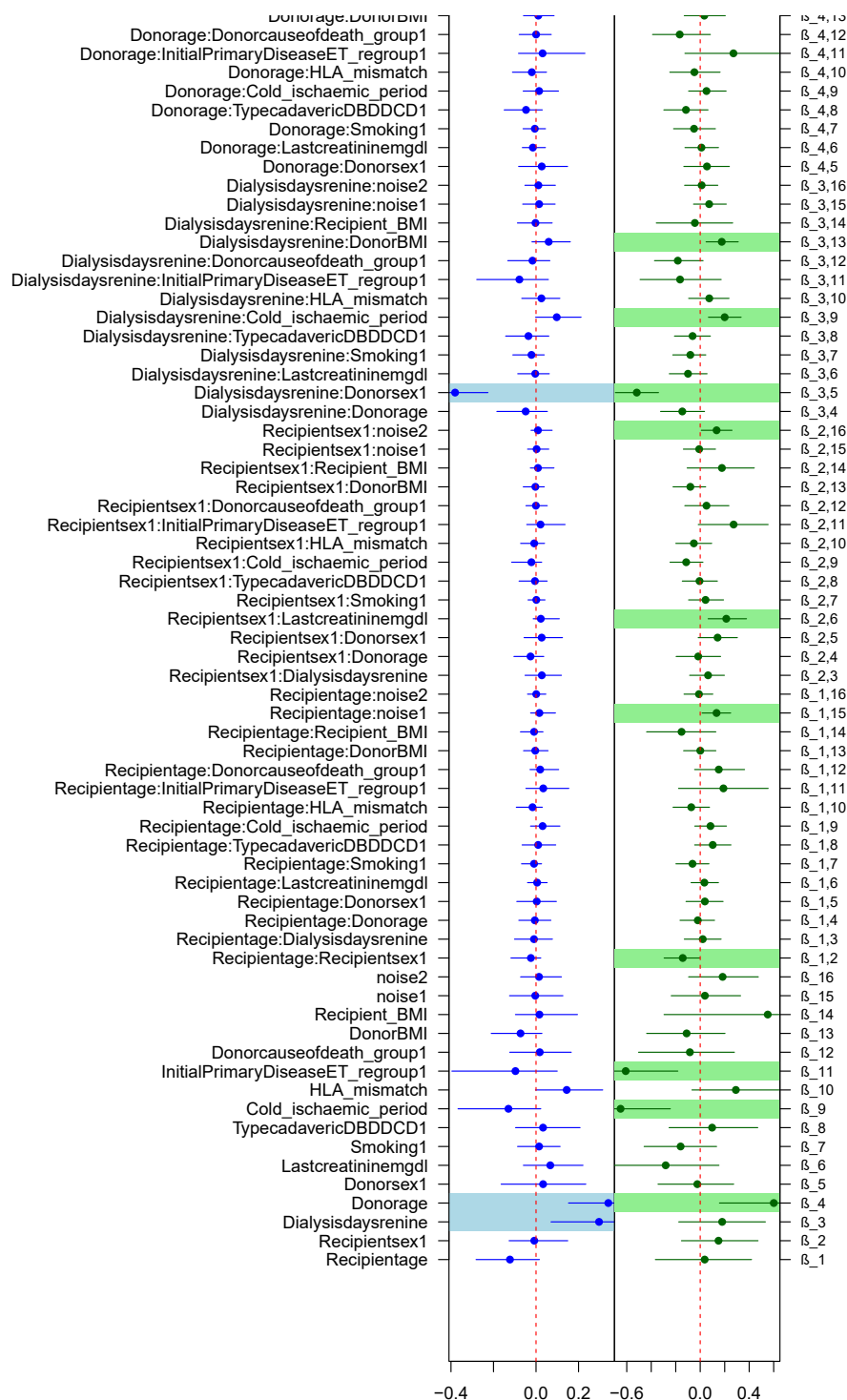


Figure 10: Results for `complexData` using `PartialLikelihood` and `PartialLikelihood.no`. Horizontal bars represent the 95% credible intervals, and shadowed terms indicate the effects selected by each method. `PartialLikelihood.no` identifies numerous covariates, including interactions with noise variables. In contrast, `PartialLikelihood` selects only `dialysesvintage`, `Donorage`, `dialysesvintage:Donorsex`, and `Donorsex:Typecadaveric`. Overall, when analyzing `complexData`, `PartialLikelihood.no` appears unreliable, and preference should be given to `PartialLikelihood`.

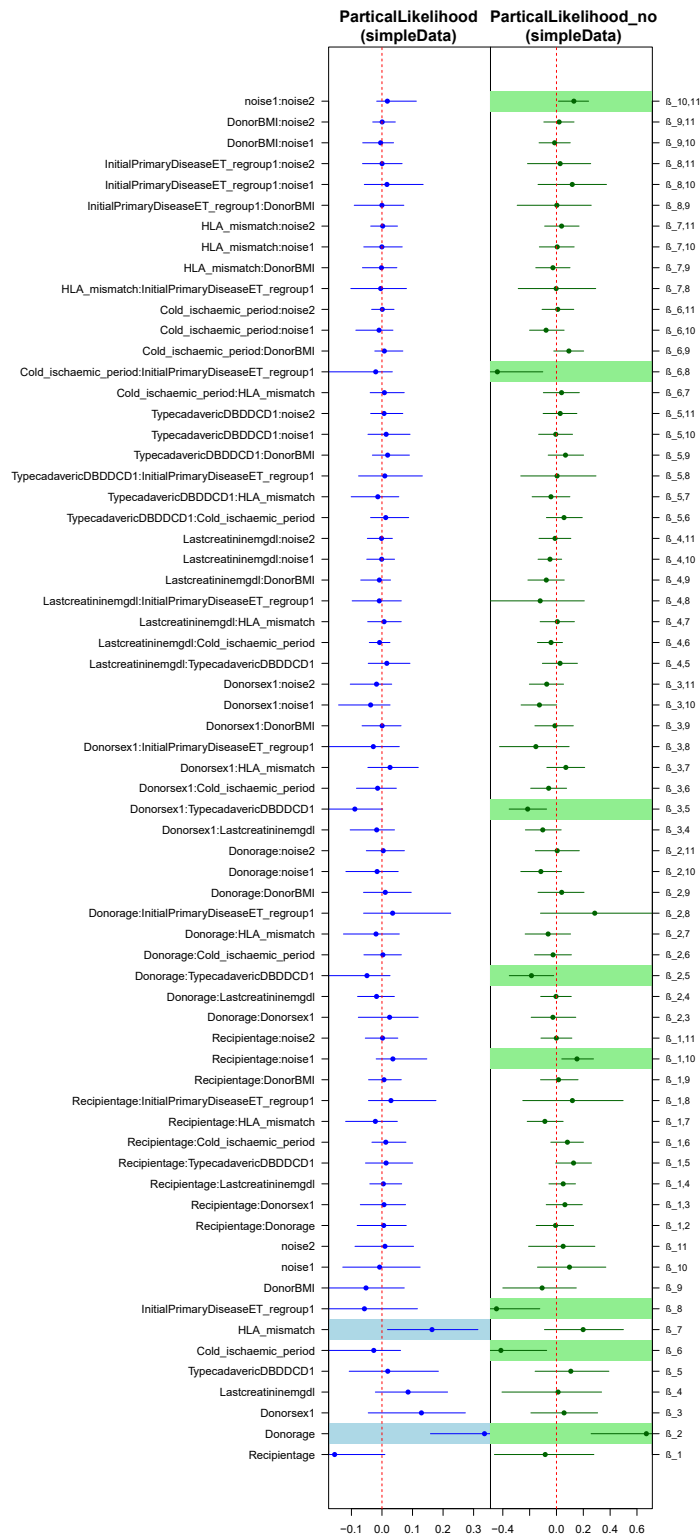


Figure 11: Results of `simpleData` from `PartialLikelihood` and `PartialLikelihood.no`. The bars represent the corresponding 95% credible intervals. Shaded terms are the selected effects by the corresponding methods. `PartialLikelihood.no` detects a lot of covariates, including the interactions with noise variable. In contrast, `PartialLikelihood` only detects the following covariates: `Donorage` and `HLA_mismatch`. Together, when analyzing `simpleData`, `PartialLikelihood.no` is not trustworthy. We should stick to the `PartialLikelihood`.

To further understand the advantages of `PartialLikelihood` over `PartialLikelihood_no`, we evaluate `PartialLikelihood` using two criteria: (1) predictive performance on kidney transplantation data, including both `complexData` and `simpleData`; and (2) correlation of posterior coefficients between the two models, considering both medians and standard deviations. As shown in Figure 12, `PartialLikelihood` achieves better predictive performance than `PartialLikelihood_no` for both `complexData` and `simpleData`. Note that predictive performance for `complexData` is better than that for `simpleData`.

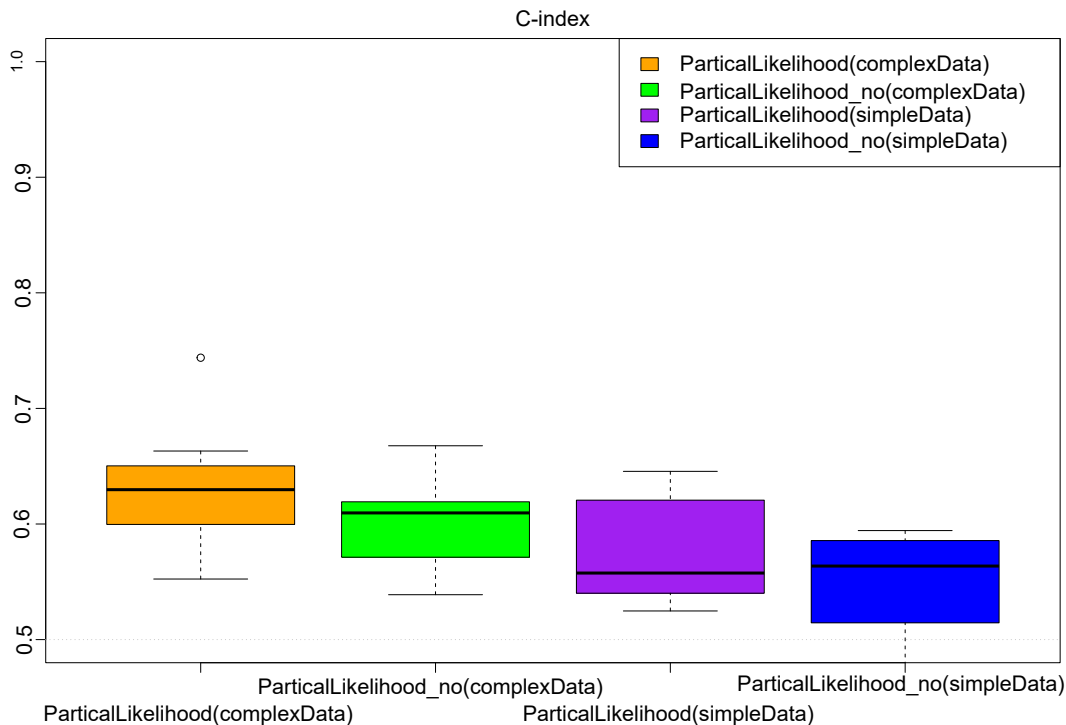


Figure 12: Predictive performance of `PartialLikelihood` and `PartialLikelihood_no` for both `complexData` and `simpleData`. Specifically, `PartialLikelihood(complexData)` and `PartialLikelihood_no(complexData)` refer to the performance of the respective methods on `complexData`, while `PartialLikelihood(simpleData)` and `PartialLikelihood_no(simpleData)` refer to their performance on `simpleData`.

Regarding correlation (Figure 13), the posterior medians are similar across `PartialLikelihood` and `PartialLikelihood_no`, regardless of the dataset. However, `PartialLikelihood` produces smaller posterior standard deviations for β compared to `PartialLikelihood_no`. This difference in posterior standard deviations is less pronounced for `simpleData` than for `complexData`.

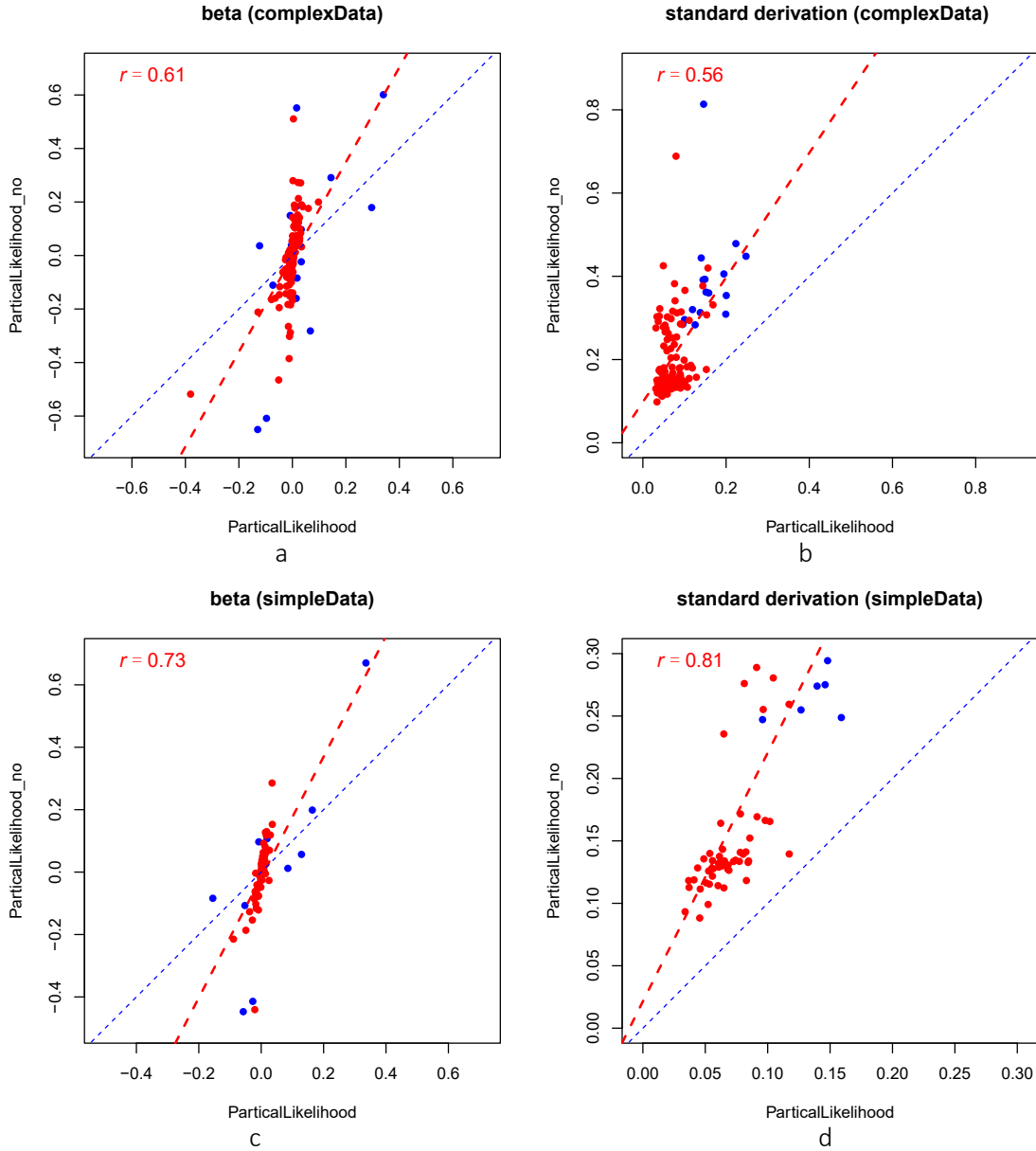


Figure 13: Correlation between posterior distributions from `PartialLikelihood` and `PartialLikelihood.no` applied to different datasets, considering both posterior medians and posterior standard deviations. The red dashed line represents the regression fit, while the blue dashed line indicates equality. Blue dots correspond to main effects, and red dots correspond to interaction effects. Panels: (a) correlation of posterior medians for `complexData`; (b) correlation of posterior standard deviations for `complexData`; (c) correlation of posterior medians for `simpleData`; and (d) correlation of posterior standard deviations for `simpleData`. Overall, `PartialLikelihood` produces similar posterior medians to `PartialLikelihood.no` across datasets, but yields smaller posterior standard deviations for β , with this difference being less pronounced for `simpleData` than for `complexData`.

In summary, Compared to `PartialLikelihood_no`, `PartialLikelihood` offers similar posterior medium but smaller posterior standard deviations, as well as better predictions. Together, `PartialLikelihood` is more trustworthy in analyzing the kidney transplantation data. Compared to `B-Splines`, `PartialLikelihood` gives more realistic estimation.

Relationship between the $\hat{\beta}$ obtained from `complexData` and `simpleData`

As noted earlier, some effects detected by `PartialLikelihood` when applied to `complexData` are not identified in the analysis of `simpleData` (see Table 5 for a summary). These include `dialysisvintage`, `HLA_mismatch`, `dialysisvintage:Donorsex`, and `Donorsex:DonorType`. Since `dialysisvintage` and `dialysisvintage:Donorsex` exist only in `complexData`, we focus on `HLA_mismatch` and `Donorsex:DonorType`.

As shown in Table 5, the credible intervals for `HLA_mismatch` and `Donorsex:DonorType` in both datasets are close to zero, suggesting that their posterior distributions are highly similar. To verify this, we computed the correlation between the posterior distributions of $\hat{\beta}$ obtained from `PartialLikelihood` applied to `complexData` and `simpleData`, considering both posterior medians and standard deviations. As illustrated in Figure 14, the posterior medians of $\hat{\beta}$ from the two datasets exhibit strong similarity, and the corresponding standard deviations follow a comparable pattern. Together, analyzing a more complex dataset with `PartialLikelihood` yields posterior medians and standard deviations similar to those obtained from a simplified dataset.

Table 5: Detected Effects by `PartialLikelihood` on `complexData` vs. `simpleData`

	PartialLikelihood(<code>complexData</code>)			PartialLikelihood(<code>simpleData</code>)		
	median	credible interval	Selected	median	credible interval	Selected
dialysisvintage	0.293	[0.071, 0.472]	Yes	-	-	-
Donorage	0.340	[0.153, 0.598]	Yes	0.336	[0.159, 0.558]	Yes
HLA_mismatch	0.144	[-0.002, 0.314]	No	0.164	[0.018, 0.314]	Yes
dialysisvintage:Donorsex	-0.129	[-0.262, -0.005]	Yes	-	-	-
Donorsex:DonorType	-0.129	[-0.262, -0.005]	Yes	-0.089	[-0.232, 0.002]	No

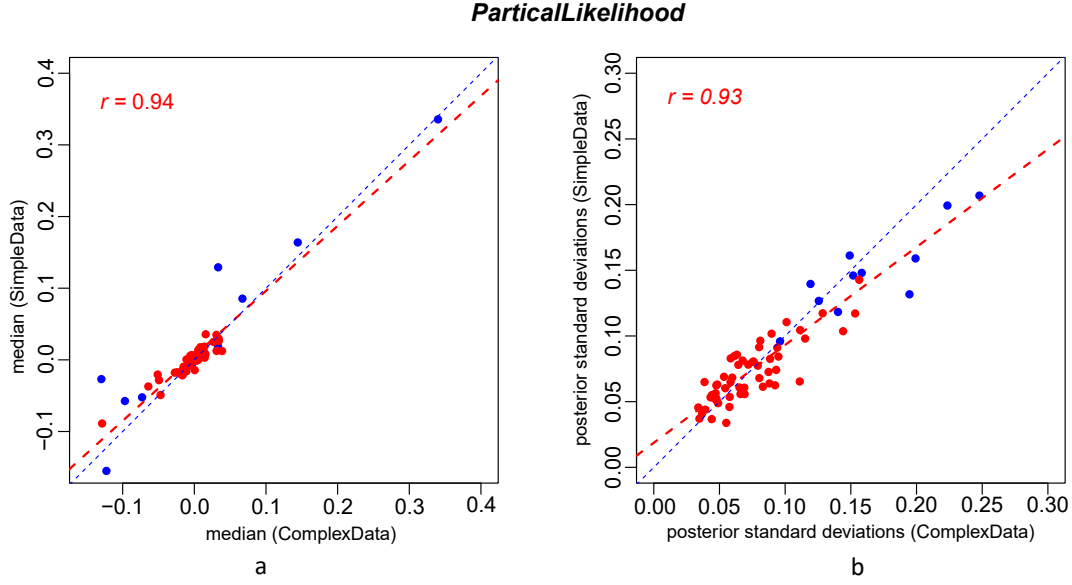


Figure 14: Correlation between posterior distributions of $\hat{\beta}$ obtained via **PartialLikelihood** from **complexData** and **simpleData**. The red dashed line indicates the regression fit, while the blue dashed line represents the identity line where the two values are equal. Blue dots correspond to main effects, and red dots represent interaction terms. Panels: (a) correlation of posterior medians; (b) correlation of posterior standard deviations. Accordingly, both posterior medians and standard deviations of $\hat{\beta}$ exhibit strong correlations, indicating high similarity derived from the two datasets via **PartialLikelihood**.

Assumption checking

As mentioned before, two key assumptions of PH models is non-informative censoring and proportional hazards. With respect to the non-informative censoring, the censored time C_i has been recorded conditionally independently of the observed time T_i given the covariates $\mathbf{x}_i$. With respect to the proportional hazards, we employ the Schoenfeld residual to check the proportional hazard assumption. As a result, we found that some variable might violate PH assumptions, i.e., **Donorage**, **Lastcreatininmgdl**, **Cold.ischaemic.period**, and **DonorBMI** (see Figure 15). This violation of the PH assumptions can also be reflected on its corresponding interactions.

Discussion

In summary, to achieve accurate estimation and reliable selection of interaction effects for better explaining graft loss in kidney transplantation, we incorporate the previously proposed linked shrinkage prior into Bayesian proportional hazards (PH) models. To this end, we employ both

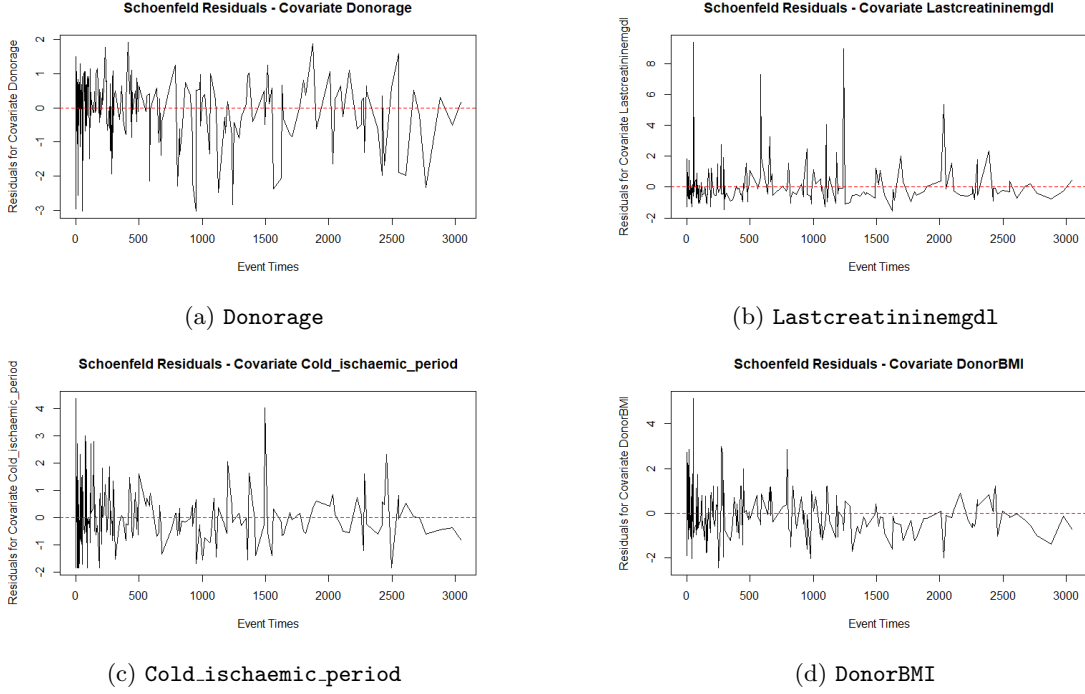


Figure 15: Results of PH assumption checking. The Schoenfeld residuals for **Donorage**, **Lastcreatininmgdl**, **Cold_ischaemic_period**, and **DonorBMI** exhibit time-dependent patterns, indicating a violation of the PH assumption for these variables.

full and partial likelihood estimators. For the full likelihood estimator, we consider three scenarios for the baseline hazard function: (1) a constant baseline hazard, where true event times T^* follow $\text{Exp}(\lambda)$; (2) an increasing baseline hazard, where T^* follows $\text{Weibull}(\lambda, v)$; and (3) an approximation of $\log h_0(t)$ using B-spline methods. For the partial likelihood estimator, the baseline hazard function is effectively avoided. The methods and their corresponding characteristics are summarized in Table 6.

Table 6: Overview of model specifications and associated characteristics

Model	Likelihood (Full vs. Partial)	Baseline Hazard $h_0(t)$	Prior for β	Ancillary Parameters and Priors
Exponential	Full	λ	Linked shrinkage	$\lambda \sim \text{Lognormal}(0, 100)$
Weibull	Full	$\lambda v t^{v-1}$	Linked shrinkage	$\lambda \sim \text{Lognormal}(0, 100)$ $v \sim \text{Lognormal}(0, 100)$
B-Spline	Full	$\exp\left(\sum_{l=1}^L \gamma_l B_l(t; k, \zeta)\right)$	Linked shrinkage	$\gamma_l \sim \mathcal{N}(0, 100)$
PartialLikelihood	Partial	—	Linked shrinkage	—
B-Spline_no	Full	$\exp\left(\sum_{l=1}^L \gamma_l B_l(t; k, \zeta)\right)$	$\mathcal{N}(0, 100)$	$\gamma_l \sim \mathcal{N}(0, 100)$
PartialLikelihood_no	Partial	—	$\mathcal{N}(0, 100)$	—

In the simulation study, we demonstrate that when the true baseline hazard function is unknown, both **B-Splines** and **PartialLikelihood** models including all two-way interactions outperform models using a non-informative prior with respect to parameter estimation, variable selection, hypothesis-testing power, and prediction accuracy. However, no consistent performance difference is observed between **B-Splines** and **PartialLikelihood**. In the analysis of real-world kidney transplantation data, the advantage of the linked shrinkage prior is evident for **PartialLikelihood** but not for **B-Splines**. Moreover, **PartialLikelihood** produces similar posterior distributions of β before and after reducing the number of covariates in the kidney transplantation dataset.

The outperformance of **B-Splines** and **PartialLikelihood** in the simulation study highlights the effectiveness of the linked shrinkage prior in improving parameter estimation, particularly for models that accounting for all two-way interactions. This finding aligns with results reported by van de Wiel et al. (2024) in regression contexts. For the real-world kidney transplantation analysis, we examine the results from two perspectives: (1) the outperformance of **PartialLikelihood** compared to the underperformance of **B-Splines**; and (2) the posterior distributions of $\hat{\beta}$.

The outperformance of PartialLikelihood compared to the weaker performance of B-Splines in kidney transplantation data

The outperformance of **PartialLikelihood** compared to the weaker performance of **B-Splines** suggests that, although the linked shrinkage prior provides clear benefits for improving parameter estimation in models incorporating all two-way interactions, its advantage becomes limited when **B-Splines** are used to approximate $\log h_0(t)$ in the kidney transplantation dataset ana-

lyzed in this study. To explore these limitations, we examine the theoretical differences between **B-Splines** and **PartialLikelihood** and identify three key distinctions in their approaches to modeling $h_0(t)$ that may explain the observed limitations of **B-Splines**.

First, **B-Splines** uses a set of knots that may lead to overfitting, unlike **PartialLikelihood**. In this study, four knots were used: two evenly spaced internal knots based on quantiles and two boundary knots, resulting in a B-spline basis with three columns. This three-column basis is unlikely to cause overfitting of the baseline hazard function $h_0(t)$. Therefore, we argue that the number of knots do not account for the weak performance of **B-Splines** compared to **PartialLikelihood** in the kidney transplantation analysis, and we discard this explanation.

Second, **B-Splines** directly estimates $h_0(t)$, whereas **PartialLikelihood** avoids modeling the baseline hazard function. The explicit modeling of $h_0(t)$ in **B-Splines** may implicitly absorb unaccounted time-varying effects (Sleeper & Harrington, 1990), such as `Cold_ischemic_period` in the kidney transplantation data. In contrast, coefficients estimated by **PartialLikelihood** represent the averaged effect for time-varying covariates over time and the actual effect for time-constant covariates. Taken together, the lack of adjustment for time-varying effects may explain the inferior performance of **B-Splines** in the kidney transplantation analysis.

Last, **B-Splines** assumes that the true baseline hazard function $h_0(t)$ is smooth, which is not required by **PartialLikelihood**. Given that both the Kaplan–Meier curve and the estimated $\hat{h}_0(t)$ from **B-Splines** exhibit a sharp decline at the beginning of the follow-up period, we argue that the true $h_0(t)$ may not be smooth and is therefore not fully captured by **B-Splines**. Thus, the lack of smoothness in the true baseline hazard could contribute to the poor performance of **B-Splines**. However, we cannot draw a definitive conclusion at this stage, and the underlying mechanism behind the failure of **B-Splines** warrants further investigation in future work.

The posterior distributions of $\hat{\beta}$

Regarding the posterior distributions of $\hat{\beta}$, we examine them from two perspectives. First, the similarity between the posterior median obtained via **PartialLikelihood** and **PartialLikelihood_no**, regardless of datasets indicates that the linked shrinkage prior does not affect the location posterior distributions. The smaller posterior standard deviations of β from **PartialLikelihood** than from **PartialLikelihood_no** suggest that the linked shrinkage prior primarily affects the dispersion of the posterior distributions. This observation is consistent with the theoretical aim of the linked shrinkage prior, which is to reduce variances by borrowing information across parameters without imposing a strong directional shift on the mean or median.

Second, the similarity between the posterior distributions of $\hat{\beta}$ obtained via **PartialLikelihood** from `complexData` and `simpleData` indicates that reducing the dimensionality of kidney transplantation data does not improve estimation stability for **PartialLikelihood**, despite the intuitive expectation that simplification enhances stability. This likely reflects the influence of the sparse prior, which is less sensitive to dimensionality. Instead, simplification in this context

introduces the risk of omitting important variables and compromising the interpretability and completeness of the model. Such omissions may lead to biased conclusions about the relationships between predictors and outcomes.

Conclusion

Linked shrinkage prior is useful for improving the detection of interaction terms in time-to-event data analysis, but they require appropriate methods to approach the baseline hazard function. Compared to the full likelihood estimator with the B-Splines method for approximating the baseline hazard function, partial likelihood estimator may be a safer choice. However, the underlying reason for this advantage of the partial likelihood estimator needs to be investigated in future studies. Furthermore, the linked shrinkage prior that incorporates the partial likelihood estimator has demonstrated reasonable robustness before and after reducing the number of covariates in the kidney transplantation data.

Limitations

As shown earlier, the proportional hazards (PH) assumption does not hold for all covariates in the kidney transplantation dataset. This is partly expected, as during the recovery process following transplantation, the effects of transplantation characteristics are often overtaken by donor and recipient-related factors. On one hand, such violations of the PH assumption imply that the current results are limited and cannot be directly interpreted to explain graft loss. On the other hand, they underscore the need to account for potential time-varying effects in future research. One approach is to introduce $\beta(t)$ for covariates that violate the PH assumption. Alternatively, in this study, restricting the analysis to graft loss occurring in the early stage—such as within one year after transplantation—could also be a reasonable strategy.

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